Chapter 13 Answers

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Problem 1

(a): Let us note that, because the Group I axioms are just the system S_1 of Chapter 7, it follows that all tautologies are provable in Peano Arithmetic (as S_1 was proved to be complete). Then, pick any formula C such that x has no free occurrences (such as a propositional formula). Then, it follows that $C \supset F(x)$ is provable (as this is a tautology if $F(x)$ is provable), and therefore so is $\forall x F(x)$ by *Group Inference Rules Rule* 2.

(b): If $F(x)$ is provable, then $\forall x F(x)$ is provable (by (a)). Hence, by $A10$, because $\forall x F(x) \supset F(t)$ is provable, so is $F(\overline{n})$ for every number n by M.P. (because every number n is a term).

(c): If $F(x, y)$ is provable, then $F(\overline{n}, y)$ is provable for every n (by (b)), and then $F(\overline{n}, \overline{m})$ is provable for every n and m (by (b)).

Problem 2

(1):
For this, we first introduce into our alphabet K the symbol v. Then, we have the following axiom schemes with the predicate Stx (for string) representing all the strings of v's:
Stv
$Stx \to Stxv$
(2):
Now we introduce the symbols (and) into K, and then Vx represents all the variables in Peano Arithmetic:
$Stx \to V(x)$
(3):

We need the following axioms for the predicate Dist (for distinct):
$stx \to sty \to Dist(x)(xy)$
$Distx, y \to disty, x$
(4):
Now we introduce the symbols 0 and $'$. Then, Num (for numeral) represents the set of Peano Numerals:
$Num0$
$Numx \to Numx'$
(5):
We now add the symbols + and $\times$ to K. Then Tx (for term) represents the set of all terms:
$Numx \to Tx$
$Vx \to Tx$
$Tx \to Ty \to T(x + y)$
$Tx \to Ty \to T(x \times y)$
$Tx \to Tx'$
(6):
We now add the symbols = and $\leq$ to K. Then Afx (for atomic formula) represents the set of all atomic formulas:
$Tx \to Ty \to Afx = y$
$Tx \to Ty \to Afx \leq y$
(7):
We now add the symbols $\sim$, $\wedge$, $\vee$, $\supset$, $\forall$ and $\exists$ into K. Then Fx represents the set of all formulas:
$Afx \to Fx$
$Fx \to F{\sim}x$
$Fx \to Fy \to F(x \wedge y)$
$Fx \to Fy \to F(x \vee y)$
$Fx \to Fy \to F(x \supset y)$
$Fw \to varx \to F\forall xw$
$Fw \to varx \to f\exists xw$:
(8):
We let $Tsub$ be the predicate representing the realtion t is a term, x is a variable, y is a term, and z is the result of substituting t for all occurences of x in y.
$Tt \to Vx \to Numy \to Tsubt, x, y, y$
$Tt \to Vx \to Tsubt, x, x, t$
$Tt \to Distx, y \to Tsubt, x, y, y$
$Tsubt, x, y, z \to Tsubt, x, y', z'$
$Tsubt, x, w, z \to Tsubt, x, v, l \to Tsubt, x, (w \times v), (z \times l)$
$Tsubt, x, w, z \to Tsubt, x, v, l \to Tsubt, x(w + v), (z + l)$
$Note$: Smullyans answer omits parenthesis (where I placed them), and this seeems to be incorrect, as he defined a term (in those cases) with parenthesis.
(9):
We let $Tfsub$ be the predicate representing the relation t is a term, x is a variable, y is a formula, and z is the result of substituting t for all free occurences of x in y:

$Tsubt, x, y, z \to Tsubt, x, w, v \to Tfsubt, x, y = w, z = v$
$Tsubt, x, y, z \to Tsubt, x, w, v \to Tfsubt, x, y \leq w, z \leq v$
$Tfsubt, x, y, z \to Tfsubt, x, w, v \to Tfsubt, x, y \wedge w, z \wedge v$
$Tfsubt, x, y, z \to Tfsubt, x, w, v \to Tfsubt, x, y \vee w, z \vee v$
$Tfsubt, x, y, z \to Tfsubt, x, w, v \to Tfsubt, x, y \supset w, z \supset v$
$Tfsubt, x, y, z \to Tfsubt, x, \sim y \sim z$
$Fy \to\to Tt \to Vx \to Tfsubt, x, \forall xy, \forall xy$
$Fy \to Tt \to Vx \to Tfsubt, x, \exists xy, \exists xy$
$Distx, y \to Tfsubt, x, w, v \to Tfsubt, x, \forall yw, \forall yv$
$Distx, y \to Tfsubt, x, w, v \to Tfsubt, x \forall yw, \forall yv$
(10):
We let $Nocf$ represent the relation x is a variable, f is a formula, and x does not occur free in f:
$Distx, y \to Tsubfy, x, f, f \to Nocfx, f$
(11):
We let Nb represent the relation x and y are variables, f is a formula, and x is not bound by y in f:
$Vx \to Vy \to Afw \to Nbx, y, w$
$Nbx, y, f_1 \to Nbx, y, f_2 \to Nbx, y, f_1 \wedge f_2$
$Nbx, y, f_1 \to Nbx, y, f_2 \to Nbx, y, f_1 \vee f_2$
$Nbx, y, f_1 \to Nbx, y, f_2 \to Nbx, y, f_1 \supset f_2$
$Nbx, y, f \to Nbx, y, \sim f$
$Nbx, y, f \to Nocfx, f \to Nbx, x, f \to Nbx, y, \forall yf$
$Nbx, y, f \to Nocfx, f \to Nbx, x, f \to Nbx, y, \exists yf$
$Nbx, y, f \to Distz, y \to Nbx, y, \forall zf$
$Nbx, y, f \to Distz, y \to Nbx, y, \exists zf$
$Note$: Smullyan's answer seems to be incorrect, as it allows for $Nbx, y, \forall y \forall xF(x)$ where $Nbx, y, \forall xF(x)$. However, this seems to infact be a formula in which x is bounded to y, as there is some G (namely $\forall xF(x)$) such that $\forall yG$ is part of F and x occurs in G. The only time quantifying y will still make x not bound to y is when x does not occur in the formula, and this is why we must have the added condition that x is not bound to x (and thus, because it does not occur freely, it cannot occur at all).
(12):
We let Ocf represent t is a term, x is a variable, f is a formula, and t is free for x in f:
$Nbx, y, f \to Ocfy, x, f$
$Nt \to Vx \to Ff \to Ocft, x, f$
$Ocft, x, f \to Ocft', x, f$
$Ocft_1, x, f \to Ocft_2, x, f \to Ocft_1 + t_2, x, f$
$Ocft_1, x, f \to Ocft_2, x, f \to Ocft_1 \times t_2, x, f$
(13):
We let the predicate Ax represent mean x is an axiom.
Group I Axiom Schemes :
$Ff \to Fg \to A(f \wedge g) \supset f$
$Ff \to Fg \to A(f \wedge g) \supset g$

$Ff \to Fg \to Fh \to A[(f \wedge g) \supset h] \supset [f \supset (g \supset h)]$
$Ff \to Fg \to Fh \to A[(f \supset g) \wedge (f \supset (g \supset h))] \supset (f \supset h)$
$Ff \to Fg \to Af \supset (f \vee g)$
$Fg \to Ff \to Ag \supset (f \vee g)$
$Ff \to Fg \to Fh \to A[(f \supset h) \wedge (g \supset h)] \supset ((f \vee g) \supset h)$
$Ff \to Fg \to A((f \supset g) \wedge (f \supset \sim g)) \supset \sim f$
$Ff \to A \sim \sim f \supset f$
Group II Axiom Schemes :
$Ocft, x, f \to Tfsubt, x, f, y \to A\forall x(f) \supset y$
$Ocft, x, f \to Tfsubt, x, f, y \to Ay \supset \exists x(f)$
Group III Axiom Schemes :
$Tx \to Ty \to A(x' = y') \supset (x = y)$
$Tx \to A \sim (x' = 0)$
$Tx \to Ty \to Tz \to A((x = y) \wedge (x = z)) \supset (y = z)$
$Tx \to Ty \to A(x = y) \supset (x' = y')$
$Tx \to Ax + 0 = x$
$Tx \to Ty \to Ax + y' = (x + y)'$
$Tx \to Ax \times 0 = 0$
$Tx \to Ty \to Ax \times y' = (x \times y) + x$
$Tfsub0, x, f, z \to Tfsubx', x, f, y \to A(z \wedge \forall x(f \supset y)) \supset \forall x(f)$
$Tx \to A(x \leq 0) \equiv (x = 0)$
$Tx \to Ty \to A(x \leq y) \equiv ((x \leq y) \vee (x \leq y'))$
$Tx \to Ty \to A(x \leq y) \vee (y \leq x)$
$Tx \to Ax = x$
$Nn \to Tfsubn, x, f, y \to Ay \supset ((x = n) \supset f))$
$Ff \to A\forall x \sim (f) \supset \sim \exists x(f)$
Note :
Smullyans addition of Vx in the mathematical induction scheme is unecessary, as $Tfsubt, x, f, z$ implies that x is a variable.
(14):
We let Px represent the set of provable formulas:
$Af \to Pf$
$Px \to Px \supset y \to Py$
$Nocfx, c \to Pc \supset f \to Pc \supset \forall x(f)$
$Nocfx, c \to Pf \supset c \to P\exists x(f) \supset c$
Note :
Smullyans addition of Fc in the last two schemes is unecessary, as this is implied by $Nocfx, c$.

Problem 3

(a):
We have already shown that F being a formula, x does not occur free in F, and x is a variable are all formally representable, and therefore the sets of the corresponding Gödel numbers are Σ_1. Hence, the following formula expresses S_0:
$F_0(y) \wedge (\forall x \leq y)(V_0(x) \supset Nocf_0(x, y))$
Well, this is equivalent to the following, with $\exists y R_1(x, y)$ expressing V_0, $\exists k R_2(w, v, k)$ expressing $Nocf_0(w, v)$ and $\exists l R_3(f, l)$ expressing $F_0(f)$, with all R_i being Σ_0 :
$\exists l \exists y \exists v(R_3(f, l) \wedge (\forall z \leq f)(R_1(x, y) \supset R_2(z, f, v))))$
which is Σ_1 by Problem 10(b) of Chapter 11 (because the above formula without the $\exists l$ is Σ_1 by Problem 10(a) of Chapter 11).
(b):
$x \in P_0$ iff $Pf_0(x) \wedge S_0(x)$
and this is Σ_1 by Problem 13 of Chapter 11.
(c):
$x \in R_0$ iff $Pf_0(nx)$ where n is the Gödel number of the negation.

Problem 4

Ω_1:
We assume that $m + n = k$. We are to show that $\overline{m} + \overline{n} = k$ is provable in Peano Arithmetic. By $A16$ and Proposition 1(b), $\overline{m} + 0 = \overline{m}$ is provable. Then, we have $(\overline{m} + 0)' = \overline{m}'$ being provable by $A15$ and M.P. Then, we also have $\overline{m} + 0' = (\overline{m} + 0)'$ being provable by A17 and Proposition 1(c), and therefore, by A14, we have $\overline{m} + 0' = m'$ being provable. We can repeat this process n times, until we eventually end up with $\overline{m} + \overline{n} = \overline{k}$, the desired result.
Ω_2:
We assume that $m \times n = k$. We are to show that $\overline{m} \times \overline{n} = \overline{k}$ is provable in Peano Arithmetic. By $A18$ and Proposition 1(b), $\overline{m} \times 0 = 0$ is provable. Then, by $A19$ and Proposition 1(c), $\overline{m} \times 0' = (\overline{m} \times 0) + \overline{m}$ is provable. We repeat the process of applying $A19$ n times, until we have proven $\overline{m} \times \overline{n} = \overline{m \times n}$
Ω_3:
We assume that $m \neq n$, and are to show that $\sim(\overline{m} = \overline{n})$ is provable in Peano Arithmetic. Let m be the number which is the greater of the two (m and n). Then $m - n = k$ for some $k > 0$. Hence, by $A13$, $\sim(\overline{k} = 0)$ is provable. By $A12$, $(\overline{k}' = 0') \supset (\overline{k} = 0)$ is provable. Hence, by propositional logic (using the fact that $\sim X \supset ((Y \supset X) \supset \sim Y)$ is provable) and M.P. we can prove that $\sim(\overline{k}' = 0')$. We can repeat this process n times, eventually ending with $\sim(\overline{m} = \overline{n})$.

Ω_4:
In the base case, $(x \leq 0) \equiv (x = 0)$ is provable by $A21$. Now assume that for some n, $(x \leq n) \equiv (x = 0 \vee ... \vee x = \overline{n})$ is provable. By $A22$, $(x \leq \overline{n}') \equiv ((x \leq n) \vee (x = n'))$. Then, because $(x \leq n) \equiv (x = 0 \vee ... \vee x = \overline{n})$ is provable, $(x \leq \overline{n}') \equiv (x = 0 \vee ... \vee x = \overline{n} \vee x = \overline{n}')$ (because the formulae $X \equiv Y$ and $Z \equiv (Y \vee W)$ imply the formula $Z \equiv (X \vee W)$ by propositional logic), thus completing the induction.
Ω_5:
This is immediate from $A23$ and Proposition 1(b).

Problem 5

Note : From this problem forward, it seems that Smullyan labels the results as general answers for any extension of (R), but in fact they rely on arithmetic axioms of PA.
We are to show that every true atomic Σ_0 sentence is provable in PA. Let us recall that an atomic Σ_0 formula is either of the form $t_1 = t_2, t_1 \leq t_2$ where t_1, t_2 are both terms. In the first case, we can prove this by mathematical induction on the number of occurences of plus, times, and the successor operation in the term (it is in fact a double induction). First, we show that for every term $t_1 = \overline{n}$ where $t_1 = n$. The base case where t_1 is a numeral is immediate from $A24$. Now, make an induction assumption on the number of plus, times, and successor symbols occuring in t. So, $t = \overline{n}$ is provable for some n. By Ω_1, $\overline{n} + \overline{k} = \overline{m}$ for every natural number k, and Ω_2 proves that $\overline{n} \times \overline{k} = \overline{m}$ for every natural number k, where m is the number that $t + \overline{k}, t \times \overline{k}$ designate, respectively, and the successor operation can be proven just by taking k to be 1 in the addition case. Then, becuase $t = \overline{n}$ is provable, we can use $A25$ to prove $t + \overline{k} = \overline{m}$ or $t \times \overline{k} = \overline{m}$, or $t' = \overline{m}$. So, we have shown that, for any term t and any natural number m, if $t = m$, then $t = \overline{m}$ is provable in Peano Arithmetic. Next, we must show that $t_1 = t_2$ for every terms t_1, t_2 such that $t_1 = t_2$. Well, by definition, $t_1 = t_2$ iff both t_1, t_2 designate the same natural number n, and hence, by the above result, both $t_1 = \overline{n}$ and $t_2 = \overline{n}$ are provable. Now, by $A14$, $((x = y) \wedge (x = x)) \supset (y = x)$ is provable, and thus, if $x = y$ is provable, so is $y = x$ (using $A24$). Hence, we have $((\overline{n} = t_1) \wedge (\overline{n} = t_2))$, and thus, by $A14$, $t_1 = t_2$ is provable. Now, we need to show that $t_1 \leq t_2$ is provable for every terms t_1, t_2, if $t_1 \leq t_2$ is true. Well, assume that $t_1 \leq t_2$. Now, because these are terms, by the above, $t_1 = \overline{n}, t_2 = \overline{m}$ are provable for some numbers n and m. There are then two cases: either $n = m$, or one of m, n is greater than the other. In the first case, $t_1 \leq m$ is provable, as $t_1 = m$ can be provable after we apply $A14$ and the fact that $m = n$ is provable, and then we use Ω_4. After this, we can use $A25$ to conclude that $t_1 \leq t_2$. Now, in the second case, assume (without loss of generality) that m is the greater numeral. Well, by Ω_4,

$t_1 \leq \overline{m} \equiv (t_1 = \overline{m} \vee ... t_1 = \overline{n} \vee ... \vee t_1 = \overline{m}$. Because $t_1 = \overline{n}$ is provable, so is $t_1 \leq \overline{m}$ (by Propositional Logic). Then, because $t_2 = \overline{m}$ is provable, $t_1 \leq t_2$ is provable by $A25$.

Problem 6

The part of this proof for the true atomic Σ_0 sentences was just Problem 5, so we only need to prove the part for the false sentences.
So let us consider a false Σ_0 sentence of the form $t_1 = t_2$ where t_1, t_2 are both terms. Well, $\overline{m} \neq \overline{n}$ for the m, n that t_1, t_2 designate respectively, is provable by Ω_3. Thus, by two applications of $A25$ (one on m, the other on n), we can prove $\sim(t_1 = t_2)$. In the other case, suppose that $t_1 \leq t_2$ is the false Σ_0 sentence. We choose m, n similarly to the first part of this problem. Because $\overline{m} \leq \overline{n}$ is false, $\overline{m} = 0, ..., \overline{m} = \overline{n}$ are all false, refutable by Ω_3, and therefore, by Propositional Logic and Ω_4, $\overline{m} \leq \overline{n}$ is refutable. Then, by two applications of $A25$, $t_1 \leq t_2$ is refutable.

Problem 7

Assume that $F(\overline{0}), F(\overline{1}), ..., F(\overline{n})$ are all provable, and we are to show that the formula $(\forall x \leq \overline{n})F(x)$ is provable in PA. Recall that $(\forall x \leq \overline{n})F(x)$ is the same as $\forall x(x \leq n \supset F(x)$. By $A25$, $F(\overline{m}) \supset ((x = \overline{m}) \supset F(x)$ is provable for any number n. Hence, by M.P., we have the formulas $(x = \overline{0}) \supset F(x), ..., (x = \overline{n}) \supset F(x)$ all being provable, and thus, by Propositional logic, so is the formula $(x = \overline{0} \vee ... \vee x = \overline{n}) \supset F(x)$. By Ω_4 and Propositional logic, this implies $x \leq \overline{n} \supset F(x)$, and thus, by Proposition 1(a), $\forall x(x \leq n \supset F(x)$ is provable.

Problem 8

We can prove that PA is Σ_0 complete by showing the every Σ_0 sentence is correctly decidable (meaning it defines itself), by using complete induction. In the base case, every atomic Σ_0 formula is correctly decidable by Problem 6. Let us make a complete induction assumption. Consider a Σ_0 sentence of degree $n+1$. If it is of the form $X \wedge Y, X \vee Y, X \supset Y, \sim X$, X and Y are both correctly decidable, obviously meaning the sentence is also correctly decidable.
Next, we must consider the other two cases, namely, those Σ_0 sentences of the form $(\forall x \leq \overline{n})F(x)$ or $(\exists x \leq \overline{n})F(x)$. In the first case, we assume that $(\forall x \leq \overline{n})F(x)$ is true. Now, this means that $F(\overline{0}), ..., F(\overline{n})$ are all true, and

thus, by the induction assumption, they are all correctly decidable, and hence provable, and thus, by Problem 7, so is $(\forall x \leq \overline{n})F(x)$. Now, assume that $(\forall x \leq \overline{n})F(x)$ is false. This implies that, for some $m \leq n$, $F(\overline{m})$ is false, and therefore, because $F(\overline{m})$ is correctly decidable by the induction assumption, it is refutable. Now, $A10$ implies $\sim F(t) \supset \sim \forall x F(x)$. Well, we now know that $\overline{m} \leq \overline{n} \wedge \sim F(\overline{m})$ is provable (because all atomic Σ_0 formulas have been shown to be correctly decidable) and thus so is $\sim(\overline{m} \leq \overline{n} \supset F(\overline{m})$, and hence, so is $\sim \forall x(x \leq \overline{n} \supset F(x)$, which is the same as saying $\sim(\forall x \leq \overline{n})F(x)$, and thus we have shown $(\forall x \leq \overline{n})F(x)$ to be correctly decidable.
On the other hand, consider $(\exists x \leq \overline{n})F(x)$. Assume that it is true. Hence, for some $m \leq n$, $F(\overline{m})$ is provable. Because $\overline{m} \leq \overline{n}$ is also provable, so is $\overline{m} \leq \overline{n} \supset F(\overline{m})$, and thus, by $A11$, so is $\exists x(\overline{x} \leq \overline{n} \supset F(x)$, which is the same as saying $(\exists x \leq \overline{n})F(x)$. Now, assume that $(\exists x \leq \overline{n})F(x)$ is false. Thus, by the induction assumption, $\sim F(\overline{0}), ..., \sim F(\overline{n})$ are all correctly decidable, true, and hence provable. This implies (by Problem 7) the sentence $(\forall x \leq \overline{n}) \sim F(x)$ which implies $\forall x(x \leq \overline{n} \supset \sim F(x)$, and thus, by $A10$, $x \leq \overline{n} \supset \sim F(x)$ is provable, which implies $\sim(x \leq \overline{n} \wedge F(x))$, and thus, by Proposition 1(a), so is $\forall x \sim(x \leq \overline{n} \wedge F(x))$, and thus, by $A26$, $\sim \exists x(x \leq \overline{n} \wedge F(x)$ is provable, which is the same as saying $\sim(\exists x \leq \overline{n})F(x)$.

Problem 9

L_1: This is immediate from Problem 7.
L_2: This is immediate from Ω_5 and Proposition 1(a).